

Daniel Mitchell

Senior Data Engineer

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📍 Melrose, MA

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SKILLS

Languages

- Python, SQL, Java, Scala, C

Data Engineering & Pipelines

- ETL / ELT Design, Apache Airflow, Apache Spark, Kafka
- Batch & Streaming Architectures, Spark Structured Streaming
- Workflow Orchestration, CDC Patterns, Data Lake Processing

Cloud & DevOps

- AWS (S3, EMR, Glue, Redshift), GCP (BigQuery)
- Docker, Kubernetes, Terraform
- CI/CD Pipelines, Infrastructure as Code

Data Warehousing & Modeling

- Snowflake, Redshift, PostgreSQL, MySQL
- Dimensional Modeling, Star/Snowflake Schema
- Data Lake & Lakehouse Architecture

Data Quality & Governance

- Data Validation Frameworks, Data Lineage
- Schema Governance, Metadata Management
- Privacy-Aware Data Design (GDPR-aligned)

EDUCATION

Master of Science, Computer Science

The City College of New York

Doctor of Philosophy, Biochemistry

Florida State University

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Meta

12/2020 – Present

- Architected the **Unified Event Analytics Platform (UEAP)** leveraging Kafka, Spark, and Airflow to enable reliable data ingestion and transformation for experimentation and personalization initiatives across core product domains.
- Redesigned end-to-end ETL workflows in Python and SQL, reducing data availability latency from several hours to under one hour and improving SLA adherence to 99.9%.
- Engineered resilient, high-throughput streaming pipelines with optimized partitioning and consumer scaling strategies to ensure consistent processing of global user interaction data.
- Advanced Spark performance by implementing data skew mitigation, memory optimization, and intelligent caching, improving job execution efficiency by 30%.
- Established governed transformation standards using dbt and dimensional modeling principles, reducing downstream data defects by 35% and improving analytical consistency.
- Modernized Airflow orchestration patterns to reliably support 5K+ concurrent tasks, strengthening scheduler stability during peak processing windows.
- Designed comprehensive observability frameworks with structured logging, metrics instrumentation, and proactive alerting, decreasing mean time to resolution by 40%.
- Collaborated closely with ML scientists and product stakeholders to develop trusted feature datasets, accelerating experimentation cycles and improving decision-making velocity.

Data Engineer

iRobot

03/2019 – 12/2020

- Built the **Connected Device Telemetry Platform (CDTP)** to centralize ingestion and processing of IoT device data, enabling actionable operational insights across global robotics deployments.
- Developed low-latency Kafka-based streaming architectures that reduced device health alerting latency by 70%.
- Implemented standardized Airflow orchestration frameworks for over 100 production pipelines, enhancing reliability and simplifying operational maintenance.
- Re-engineered warehouse schema design and query optimization strategies, improving analytical dashboard responsiveness by 30%.
- Partnered with firmware, product, and analytics teams to formalize data contracts and schema governance, minimizing cross-team integration issues.

Software Engineer, Data

JW Player

01/2017 – 03/2019

- Developed the **Video Analytics & Ad Insights Platform**, delivering scalable backend data services for engagement tracking and advertising performance measurement.
- Designed distributed aggregation and enrichment pipelines that reduced reporting turnaround time for enterprise media clients.
- Strengthened platform resilience through enhanced validation logic, monitoring instrumentation, and incident response processes.